

MATTHEW ZHOU

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EDUCATION

Georgia Institute of Technology

[May 2027 (Expected)]

- **B.S. Mechanical Engineering**, Concentration in Automation & Robotics, Minor in German (GPA: 3.97/4.00).
- Relevant Coursework: Robotics, Machine Design, Design & Manufacturing, Heat Transfer, ME Systems Laboratory

EXPERIENCE

RoboRambler GT (ARCC/RoboMaster Competition Team) | Co-Founder, Ex-President & Mechanical Lead [September 2023 – Present]

Built two generations of FPV mobile combat robots with artillery, advanced from bottom-ranked to podium finish in 3 years

- Finished 2nd in 1v1 competition and 9th in 3v3 competition among over 30 teams worldwide during ARCC 2026 (formerly RoboMaster North America); finished 15th in 1v1 competition and 17th in 3v3 competition in RoboMaster NA 2025.
- Obtained workspace in Tech Square ATL in 2025 and helped start Humanoids @ GT club to diversify student robotics endeavors.
- Developed X-drive chassis with suspension, eliminating off-road FPV blur and traction loss due to aggressive maneuver, featuring custom omni-wheel design with carbon fiber plates and molded TPU roller, achieving 20% less weight than stock wheels.
- Designed gun weight compensation mechanism with trigonometry, consuming near 0 power to aim a 3-kg gun at any pitch angle.
- Used heat-treated PA6-CF prints, pocketed CF plates, EDM-machined aluminum extrusions, and CNC metal gun chambers for 2026 fleet, coordinated tolerancing, finishing, and logistics with international manufacturers, making robots 8% lighter than 2025 fleet.
- Developed 20-RPS 17mm-caliber flywheel artillery and 42mm-caliber with <0.1 m dispersion at 5 m with custom curved flywheels.
- Secured donated through-bore slip rings, enabling infinite turret rotation on chassis with reliable electrical connection and ammo supply; led talks with companies for spatial memory unit, 3D printing filament, and humanoid actuator donations worth \$7,000+.

Volkswagen AG | Assembly Maintenance Engineering Intern at Chattanooga Plant

[May 2026 – Present]

Providing improvement solutions to VW Atlas production lines and saving time and costs

- Designed mechanical-driven push skid system sweeper with helical brush, reducing faults and estimated to save 5 lost cars monthly.
- Upgraded an old glue cell's robot gripper and sensor configuration to accommodate VW 2027 Atlas quarter glass as back-up, saving the company over \$90,000 outsourcing cost, 6 weeks of lead time, and 40 minutes of down time on average every week.
- Designed a lightweight welded-steel platform to bridge a 0.6 m gap between an elevator and ladder at 15 m elevation, providing safe standing access for one maintenance personnel and improving OSHA compliance.

Advanced Manufacturing Pilot Facility (GT Manufacturing Institute) | Research Assistant

[January 2024 – Present]

Worked on projects that facilitate mechanical testing automation and support advanced manufacturing techniques

- Received President's Undergraduate Research Award for a study adapting twin bridge geometry to investigate anisotropic behaviors in circular metal extrusions, determined testable & machinable sample dimension based on material property, load and grip capacity.
- Created and FEA-validated fixture for the designed sample, compatible with torsional and axial load case and DIC mirror mounting.
- Strain-validated an induction-heated printing buildplate with 60 lb. print capability with hand-soldered strain gauges and FEA.
- Certified as student coach on Wire EDM with 100+ hours of experience, trained peers, and machined alloy testing samples.
- Presented facility Directed Energy Deposition air supply analysis at GT Mfg. 4.0 Consortium, showcasing scheduling strategies.

Marine Robotics Group at Georgia Tech | Mechanical Lead for RoboSub Team

[September 2023 – May 2025]

Developed a fully autonomous vehicle performing maneuver and manipulation tasks underwater

- Co-designed main vehicle hull and thruster configurations that empower the vehicle with 6-DOF underwater maneuverability.
- Engineered gripper, external battery tube, torpedo launcher, and sensor mounts to accomplish underwater manipulation tasks.
- Developed a neutrally buoyant autonomous torpedo with 2 rear thrusters, capable of self-guiding to a 4-in-diameter target from 3 ft.

SKILLS

Software: SolidWorks ([SOLIDWORKS Design Professional \(CSWP\)](#)), CATIA V5, Onshape, Microsoft Office Suite.

Programming: MATLAB, Arduino, Java.

Languages: English, German, Chinese (Mandarin), Chinese (Cantonese)

Others: 3D Printing & Heat Treatment, Project Management, Corporate Sponsorship Negotiation, Soldering, Wire EDM, Mill, Lathe, FEA.